Global Olympic Sports Performance Analysis

Project Report: Data Analytics with Tableau

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1. Introduction

1.1 Project Overview

The Olympic Games bring together many countries, sports and athletes, and the data from them is large and spread across many Games. This project analyses four project datasets (athletes, medal results, Games summary and all-time country rankings) covering 18 Olympic Games from 1992 to 2026, 60 countries, 50 sports and 7 regions. The data is presented through two interactive Tableau dashboards, a Tableau story and a Flask website.

- **Dashboard 1 (Global Sports Performance Analytics):** KPI cards (global athlete participation, total sports categories entered, total nations represented), a world map of total medal achievements across nations, a line chart of total medals awarded over the years, average medals per athlete by sport category, and global performance metrics by country with a Total / Gold / Silver / Bronze selector.
- **Dashboard 2 (Olympic Performance and Regional Efficiency):** regional KPIs (regional performance efficiency, countries participating, athlete participation count, average medal efficiency), average medal efficiency percentage by region, global sports investment vs medal performance, medal totals across Olympic hosts, and regional medal efficiency and distribution by sport.

Tableau Public link: (add your Tableau Public link here after publishing)

1.2 Purpose

The purpose is to turn raw Olympic data into clear, interactive insights that help sports strategists, investment analysts and journalists understand which sports and countries win the most medals, how medals changed over the years, which regions are the most efficient, and how host nations performed, so they can make informed decisions and tell clear stories.

2. Ideation Phase

2.1 Problem Statement

Olympic data is large and spread across many files, Games and countries. Without one consolidated view, it is hard to see which sports, countries and regions perform best, how medals relate to sports investment, and where medals are won most efficiently. Three stakeholders need tailored answers:

Persona	Role	Need
Ms. Priya	Performance Strategist, National Olympic Committee	Which sport categories give the most medals per athlete, and how her country compares with top nations, so she can recommend where to invest before the next Olympic cycle.
Mr. Carlos	Sports Investment Analyst, International Funding Agency	How sports investment relates to medal performance, which nations and regions are efficient, and which regions deserve more support when allocating grants.
William Robert	Broadcast Journalist and Data Storyteller, Olympic Media Network	How total medals changed over the years, how host nations performed, and how regions are distributed by sport, to build a documentary story.

2.2 Empathy Map Canvas

Quadrant	Priya (Strategist)	Carlos (Investment Analyst)	William (Journalist)
Says	"Which sports should we invest in?"	"Who gets the best medal return?"	"What is the story across the Games?"
Thinks	Needs fast, evidence-based views to back her recommendation	Grants must go where they make a difference	Viewers need clear, data-backed narratives
Does	Compares sport categories and top countries	Compares investment, efficiency and regions	Studies medal trends, hosts and regional patterns
Feels	Pressure to choose the right sports	Needs confidence before allocating funds	Motivated to tell an engaging story

2.3 Brainstorming

Ideas considered and grouped by theme:

- **Global Medal Performance:** KPI cards, world medal map, medal trend over the Games, top countries with a Total / Gold / Silver / Bronze selector.
- **Efficiency and Investment:** average medal efficiency by region, sports investment vs medal performance, regional distribution by sport.
- **Host and Sport Analysis:** medal totals across Olympic hosts and average medals per athlete by sport category.
- **Interactivity and Storytelling:** filters (Country, Year, Sport Category, Region, Host Country), a Tableau story and a Flask website.
- **Selected:** two dashboards (Global Sports Performance Analytics, Olympic Performance and Regional Efficiency) joined by a Tableau story and shown on a Flask website.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User action	Dashboard support
Awareness	Opens the dashboard to see the Olympic picture at a glance	KPI cards, world map and medal trend on Dashboard 1
Exploration	Filters by country, year, sport category or region and compares charts	Filters, KPI Selector and tooltips
Analysis	Digs into medals per athlete, efficiency, investment and hosts	Bar, pie, bubble and donut charts on both dashboards
Decision	Picks sports and regions to support	Efficiency, investment and sport-by-region views
Sharing	Presents findings to committees, funders or viewers	Tableau story, Tableau Public link and Flask website

3.2 Solution Requirement

Functional requirements

- Connect the four CSV files and relate them on noc_code and year.
- Display headline KPIs: athlete participation, sports categories entered, nations represented and regional efficiency.
- Show medals by country on a world map, and total medals awarded over the years.

- Compare average medals per athlete by sport category, and country performance with a Total / Gold / Silver / Bronze selector.
- Compare sports investment with medal performance, and average medal efficiency by region.
- Compare medal totals across Olympic hosts, and regional medal distribution by sport.
- Provide interactive filtering by Country, Year, Sport Category, Region and Host Country.

Non-functional requirements

- Clear, readable layout with titles, labels and consistent colours.
- Fast loading and responsive filters on small CSV files.
- Publicly shareable through Tableau Public and a Flask website.
- No personal or confidential data is stored or published.

3.3 Data Flow Diagram

Four CSV files (athletes, medal_results, games_summary, alltime_country_rankings) → Tableau data source connection → relationships on noc_code and year, data checks and calculated fields (Year, KPI Selector) → worksheets (charts) → Dashboard 1 and Dashboard 2 → Story → Tableau Public → Flask website.

3.4 Technology Stack

Layer	Tool
Data source	Four CSV files provided with the project
Data preparation	Tableau relationships, data checks, calculated fields and a KPI Selector parameter
Visualisation	Tableau Desktop (Public Edition)
Publishing	Tableau Public
Web front-end	Flask (Python), HTML and CSS
Version control	GitHub
Documentation	PDF and Word documents

4. Project Design

4.1 Problem Solution Fit

The problem is scattered Olympic data with no single view for decision makers. The solution is a pair of focused interactive dashboards and a story that answer each persona's questions directly, so insights that would take hours in raw tables are available on one screen.

4.2 Proposed Solution

- Dashboard 1: KPI cards, world medal map, medal trend, average medals per athlete by sport category, country performance.
- Dashboard 2: regional KPIs, efficiency by region, investment vs medal performance, host medals, regional distribution by sport.
- Story: a guided sequence of both dashboards with key observations.
- Delivery: published on Tableau Public and shown on a Flask website, with files stored on GitHub.

4.3 Solution Architecture

Data layer (four CSV files) → Tableau data model with relationships and calculated fields → worksheet layer (charts) → dashboard layer (Dashboard 1 and Dashboard 2) → story layer → Tableau Public and Flask website for end users (Priya, Carlos and William).

5. Project Planning & Scheduling

5.1 Project Planning

Phase	Activities	Output
1. Understanding	Study the problem statement, scenarios and personas	Business questions
2. Data preparation	Load the four CSV files, relate them, check fields, create calculated fields	Tableau data source
3. Visualisation	Build worksheets for each business question	Worksheets
4. Dashboards	Combine worksheets into two dashboards with filters	Dashboard 1 and 2
5. Story and publishing	Build the story and publish to Tableau Public	Public link
6. Web integration	Build the Flask website and embed the dashboards and story	Flask website
7. Documentation	Complete the report, templates and demo video	Final report and video

6. Functional and Performance Testing

6.1 Performance Testing

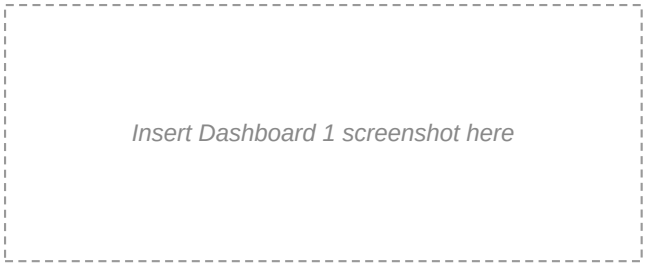
Update the Status column only after each test has really been done.

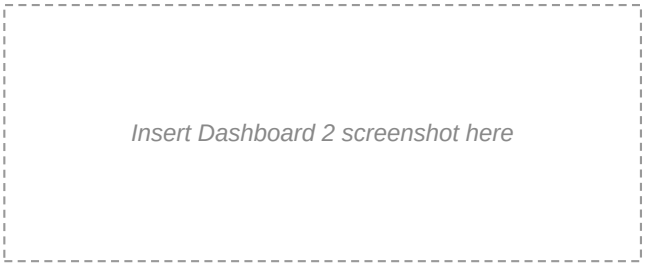
Test	Expected result	Status
Data connection	All four files load and relate on noc_code and year without errors	Pending
KPI cards display correct totals	60 nations and 21 sport categories are shown; athlete totals match the data	Pending
KPI Selector	The country chart switches between Total, Gold, Silver and Bronze	Pending
Filters (Country, Year, Sport Category, Region, Host Country)	Charts update for the selected values	Pending
Tooltips on charts	Values are shown on hover	Pending
Dashboards on Tableau Public	Both dashboards and the story render fully	Pending
Flask website	Home, Dashboard and Story pages open and show the embedded views	Pending

7. Results

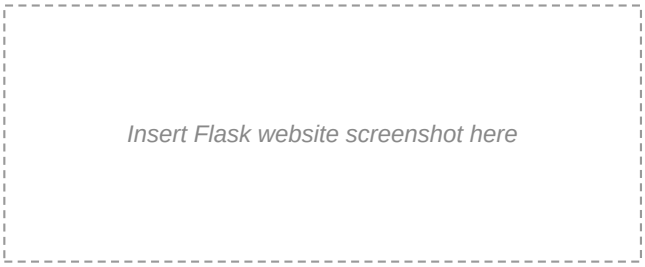
7.1 Output Screenshots

Insert the Dashboard 1, Dashboard 2, story and website screenshots here.









Key findings

These values come from the four CSV files. Check each one against your own dashboards before you submit.

- The data covers 8,000 athlete records, 60 countries, 50 sports in 21 sport categories, and 18 Games from 1992 to 2026 (9 Summer and 9 Winter).
- Medals awarded per Games ranged from 305 in 2000 to 363 in 2020, and 334 were awarded in 2026.
- In the all-time rankings the United States leads with 735 medals (250 Gold), followed by China with 696 and Russia with 429. Great Britain (401) and Germany (336) complete the top five.
- Europe holds the largest share of all-time medals (about 47%), followed by Asia (about 25%) and North America (about 17%).
- South America has the highest average medal efficiency (49.64%), followed by Europe/Asia (45.98%) and Africa (42.80%). North America has the lowest (27.46%).
- Countries with High sports investment won on average about 18.7 medals per Games, compared with about 3.4 for Medium and 2.4 for Low investment, although Low-investment countries show a higher efficiency percentage.
- Biathlon has the highest average medals per athlete among sport categories (1.69), followed by Shooting (1.66) and Cycling (1.62). Sliding is the lowest (1.31).
- Host nations won many medals on home ground, for example China with 63 in 2008 and 55 in 2022, and the United States with 58 in both 1996 and 2002.

8. Advantages & Disadvantages

Advantages

- Single interactive view of medals, participation, regions, investment and hosts.
- Serves several personas with one project.
- Easy to share and explore through Tableau Public and a website.
- Supports data-driven investment and storytelling decisions.

Disadvantages

- Depends on the quality and coverage of the four datasets.
- Tableau Public data is publicly visible.
- Sports investment is given as High, Medium and Low levels, not exact amounts.
- Data is a snapshot and does not update in real time.

9. Conclusion

The project shows how Tableau can convert Olympic data into actionable insight. The dashboards reveal which countries and sport categories lead, how medals changed across the Games, which regions are the most efficient and how host nations performed. Together they give strategists, funders and journalists a clear basis for planning, allocation and storytelling.

10. Future Scope

- Connect live or regularly refreshed data sources.
- Add forecasting of medals for upcoming Games.
- Include actual investment amounts and population or GDP trends.
- Add country-level drill-downs and comparison views.
- Build mobile-optimised dashboard layouts.

11. Appendix

Source Code: Flask website files (app.py and templates/index.html) are in the GitHub repository; the Tableau workbook is also included.

Dataset Link: used four local .csv files provided with the project (athletes, medal_results, games_summary, alltime_country_rankings).

GitHub: https://github.com/nandinisethiya/global-olympic-sports-performance-analysis

Demo: *(add your demo video or Tableau Public link here)*